RAG System with Retrieval Benchmarking

AI Intern Hackathon — Option B Technical Report

Comparative evaluation on TREC-COVID: two chunking strategies, two embedding models, vector vs. hybrid search, LLM reranking, and query rewriting evaluated against official NIST relevance judgements

Dataset Description and Source

TREC-COVID (BEIR Benchmark)

We use **TREC-COVID** from the BEIR retrieval benchmark — a standard corpus in academic information retrieval research consisting of biomedical papers about COVID-19 from the CORD-19 collection, with 50 queries and relevance judgements produced by trained NIST human assessors.

Property	Value
Full corpus	171,332 documents (CORD-19, Allen Institute for AI)
Queries	50 official TREC topics
Relevance judgements	Human-assessed by NIST; graded 0 = not relevant, 1 = partially relevant, 2 = highly relevant
Domain	Biomedical research on COVID-19 / SARS-CoV-2
Download	HuggingFace `BeIR/trec-covid` via `prepare_dataset.py`

Table 1: TREC-COVID dataset properties.

Subset Construction

Embedding all 171K documents would cost >$13 per config at current OpenAI pricing. We construct a focused, challenging subset:

- **800 relevant docs**: documents rated relevant (≥ 1) in qrels for any of the 50 queries, randomly sampled with `seed=42`
- **500 noise docs**: randomly selected non-relevant documents, ensuring retrieval must discriminate rather than trivially return everything
- **Result**: ∼1,300 documents → 887–1,019 chunks (varies by strategy)

Note on recall figures. R@5 is computed with the denominator being all relevant documents in the *full corpus* (average 27 per query), not just those in our subset. Absolute recall values are therefore conservative by design; a perfect retrieval of 5 results from 27 relevant documents yields R@5 $= 5/27 \approx 0.19$. **Relative differences between configurations are the meaningful signal.**

Why TREC-COVID? Unlike self-defined ground truth, qrels are produced by independent human experts, making our metrics directly comparable to published IR literature. The biomedical domain provides genuine retrieval difficulty: many documents discuss overlapping topics (cytokines, ACE inhibitors, transmission routes), requiring precise ranking rather than coarse topic matching.

System Architecture and Key Design Decisions

Component Overview

File	Responsibility
`prepare_dataset.py`	Downloads TREC-COVID from HuggingFace; constructs subset; saves corpus, queries, qrels
`rag_system.py`	Core engine: chunkers, embedding engine, FAISS index, BM25 index, hybrid fusion, reranker, query rewriter, answer generator, RAGPipeline orchestrator (565 lines)
`benchmark.py`	Evaluation framework: metric functions, official qrels loading, benchmark runner, comparison table, cost estimator
`run.py`	Entry point: loads dataset, defines 6 configurations, runs benchmarks, smart result merging

Table 2: Four-file architecture with clean interfaces.

Pipeline

Setup (once):
`prepare_dataset.py` → HuggingFace download → corpus subset + qrels saved to `data/trec_covid/`

Ingest (per config):
corpus JSON $\xrightarrow{\text{Chunker}}$ chunks $\xrightarrow{\text{OpenAI embed}}$ float32 vectors → FAISS IndexFlatIP + BM25Okapi index

Query (per question):
question $\xrightarrow{\text{(opt.) QueryRewriter/gpt-4o-mini}}$ expanded query $\xrightarrow{\text{embed}}$ query vector
→ FAISS top-10 + BM25 top-10 $\xrightarrow{\text{RRF fusion}}$ ranked results $\xrightarrow{\text{(opt.) LLMReranker}}$ top-5
→ `gpt-4o-mini` grounded answer with source citations → evaluate against official qrels

Key Design Decisions

FAISS `IndexFlatIP` + L2 normalisation. All embeddings are normalised to unit length before insertion, making inner product equivalent to cosine similarity. This gives exact nearest-neighbour search with zero approximation error, appropriate for our ∼900–1,000 chunk corpus.

Reciprocal Rank Fusion (RRF) for hybrid search. Cosine similarity scores and BM25 scores are incomparable in scale. Rather than normalising them (fragile, domain-dependent), RRF uses only rank positions: $\text{score}(d) = \sum_r \frac{1}{60+\text{rank}_r(d)}$. Robust, calibration-free, and empirically strong across IR benchmarks.

PDF-aware sentence chunking. PDF extraction produces single newlines as line-wrap artefacts, not sentence boundaries. The `SentenceChunker` first collapses single `\n`→space (protecting `\n\n` paragraph breaks), then splits on sentence-ending punctuation. Without this step, chunking generates ∼10× too many fragments (>10,000 vs ∼900).

RERANK_TOP_N = 5. The reranker returns top-5 passages after re-scoring. An initial value of 3 mechanically reduced P@5 and truncated answer context; raising to 5 improved P@5 from 0.46 to 0.82.

Token-safe batching with retry. Embedding API calls are bounded at 200K estimated tokens and 500 items per request. Failed batches retry with exponential backoff (2s→4s→8s→16s), falling back to single-item mode. This handles the null bytes, broken Unicode, and control characters common in PDF-extracted text that cause OpenAI 500 errors.

Embedding cache. All `EmbeddingEngine` calls are cached by sanitised text content. Running 6 configurations never re-embeds the same chunk twice, reducing API cost by ∼5× across a full benchmark run.

Official qrels evaluation. P@K, Recall@K, MRR, and NDCG@10 are computed against NIST-assessed relevance judgements — the same evaluation protocol used in TREC shared tasks.

Chunking Strategies Compared

	Fixed-Size (Strategy 1)	**Sentence-Boundary (Strategy 2)**
Parameters	300 words per chunk, 50-word overlap	300-word budget, 1-sentence overlap
Chunks produced	1,019	887
Method	Whitespace tokenise; slide window	Normalise PDF newlines; group sentences
Pros	Uniform size; predictable embedding quality	Semantically coherent; complete thoughts
Cons	Cuts mid-sentence	Variable chunk size; requires PDF preprocessing
Best MRR achieved	0.950	**1.000**

Table 3: Chunking strategy comparison. Sentence chunking produces 13% fewer chunks and achieves perfect MRR.

Evaluation Results

Test Queries

We select the 10 TREC-COVID queries with the highest number of relevant documents in our subset, maximising evaluation signal per query.

ID	**Official TREC Query**	**Relevant docs**
Q38	What is the mechanism of inflammatory response / pathogenesis of COVID-19?	41
Q39	What is the mechanism of cytokine storm syndrome in COVID-19?	33
Q20	Are patients taking ACE inhibitors at increased risk for COVID-19?	26
Q18	What are the best masks for preventing COVID-19 infection?	25
Q26	What are the initial symptoms of COVID-19?	25
Q6	What types of rapid testing for COVID-19 have been developed?	24
Q13	What are the transmission routes of coronavirus?	24
Q17	Are there any clinical trials available for coronavirus?	24
Q27	What is known about asymptomatic COVID-19 infections?	24
Q45	How has the COVID-19 pandemic impacted mental health?	21
Average relevant docs per query		**27**

Table 4: 10 test queries from TREC-COVID with official NIST ground truth.

Metric Definitions

- **P@K**: Fraction of top-K retrieved chunks from a document rated relevant (≥ 1) by NIST assessors.
- **R@5**: Fraction of all corpus-relevant documents found in top-5 results. Conservative by design (see Section 1.2).
- **MRR**: Mean Reciprocal Rank — $\frac{1}{N}\sum_i \frac{1}{\text{rank of first relevant result}_i}$.
- **NDCG@10**: Normalised Discounted Cumulative Gain at 10, using graded relevance (0/1/2). Rewards ranking highly-relevant docs above partially-relevant ones. Standard metric in TREC evaluations. $\text{NDCG@K} = \frac{\text{DCG@K}}{\text{IDCG@K}}$, where $\text{DCG@K} = \sum_{i=1}^{K} \frac{2^{r_i}-1}{\log_2(i+1)}$.
- **LLM score**: `gpt-4o-mini` rates each answer 0–10 on relevance, factual accuracy, and biomedical specificity; normalised to 0–1.

Main Results Table

Configuration	P@3	P@5	R@5	MRR	NDCG@10	LLM	Lat (s)	Chunks	Cost ($)
Fixed + Small + Vector	**0.933**	0.880	0.161	0.950	**0.710**	0.83	4.12	1019	0.01127
Fixed + Small + Hybrid	**0.933**	0.860	0.165	0.950	**0.714**	0.83	**3.85**	1019	0.01116
Fixed + Large + Vector	0.833	0.880	0.163	0.933	0.683	0.82	4.22	1019	0.04485
Sentence + Small + Vector	**0.933**	**0.900**	**0.173**	**1.000**	0.705	**0.84**	4.22	887	0.01058
Sentence + Small + Hybrid	**0.933**	0.860	0.165	0.950	0.708	**0.84**	4.71	887	**0.01043**
Sent.+Large+Hybrid+Rerank	0.800	0.820	0.159	0.950	0.470	0.82	11.44	887	0.03770

Table 5: Full benchmark results: 6 configurations, 10 TREC-COVID queries, official NIST qrels. Green = Pareto-optimal configs (best retrieval per cost tier). Blue = reranker config (discussed in Section 4). **Bold** = best value per column.

Per-Query Breakdown — Baseline vs. Best (Fixed+Small+Vector)

ID	Query	P@3	R@5	NDCG@10	LLM
Q38	Inflammatory response / pathogenesis	1.000	0.122	0.931	0.8
Q39	Cytokine storm mechanism	1.000	0.152	0.730	0.8
Q20	ACE inhibitors & COVID-19 risk	1.000	0.154	0.880	1.0
Q18	Best masks for prevention	1.000	0.200	0.779	0.9
Q26	Initial symptoms	1.000	0.120	0.655	0.8
Q6	Rapid testing types	0.667	0.167	0.826	0.8
Q13	Transmission routes	0.667	0.125	0.316	0.8
Q17	Clinical trials	1.000	0.167	0.626	0.8
Q27	Asymptomatic infections	1.000	0.167	0.767	0.8
Q45	Pandemic mental health impact	1.000	0.238	0.589	0.8
Average		**0.933**	**0.161**	**0.710**	**0.83**

Table 6: Per-query results for Fixed+Small+Vector (best NDCG@10 config). Orange = Q13 anomaly discussed in Section 4.

Reranker Before vs. After RERANK_TOP_N Fix

Config	P@3	P@5	R@5	MRR	NDCG@10	LLM
Sent.+Large+Hybrid+Rerank (TOP_N=3, initial run)	0.767	0.460	0.089	0.850	0.327	0.81
Sent.+Large+Hybrid+Rerank (TOP_N=5, current)	0.800	0.820	0.159	0.950	0.470	0.82

Table 7: Impact of raising RERANK_TOP_N from 3 to 5. P@5 improved from 0.460 to 0.820 (+78%), R@5 from 0.089 to 0.159 (+79%), NDCG from 0.327 to 0.470 (+44%). This single parameter change demonstrates the importance of returning sufficient context for both evaluation and answer generation.

Discussion of Trade-offs, Hallucination Cases, and Improvements

Key Trade-off Findings

Finding 1: Small embeddings outperform large on this biomedical corpus. Fixed+Small+Vector (NDCG = 0.710) outperforms Fixed+Large+Vector (0.683) at 4× lower cost ($0.011 vs $0.045 per run). The 3072-dimensional `text-embedding-3-large` model performs worse despite being more expensive. The likely mechanism: biomedical queries have strong lexical overlap with relevant documents — "cytokine storm" appears verbatim in both query and paper titles. Larger embeddings

over-generalise semantically, retrieving topically related but imprecise documents. **Practical implication**: always benchmark embedding model size before upgrading; larger is not always better.

Finding 2: Sentence chunking achieves perfect MRR at lower cost. Sentence+Small+Vector is the only configuration with MRR = 1.000, meaning the first retrieved result was always from a relevant document. It also achieves the highest P@5 (0.900) and R@5 (0.173), and costs 6% less than the Fixed+Small equivalent ($0.01058 vs $0.01127). Sentence-boundary chunks preserve complete clinical claims, producing better query-chunk alignment for biomedical questions where a single well-formed sentence often contains the full answer.

Finding 3: Hybrid search is query-type dependent, not universally better. Comparing Fixed+Small+Vector (NDCG = 0.710) vs Fixed+Small+Hybrid (0.714): only a 0.4-point improvement on average. However, the per-query variance is significant: hybrid improves P@3 on Q6 ("rapid testing": 0.667→1.000) because BM25 rewards the exact phrase match; hybrid hurts Q26 ("initial symptoms": MRR 1.0→0.5) because BM25 promotes documents containing the word "symptoms" in unrelated clinical contexts. This is the canonical hybrid retrieval trade-off: BM25 is a precision tool for lexically specific queries, not a general improvement.

Finding 4: Reranker quality is highly sensitive to context window size. Raising RERANK_TOP_N from 3 to 5 improved P@5 by 78% and NDCG by 44%. This reveals that the initial poor reranker performance was not a reranking quality problem but a context truncation problem: with only 3 passages, both evaluation metrics and generation context were insufficient. Even with the fix, the reranker (NDCG = 0.470) still underperforms simple baselines (NDCG = 0.710). The remaining gap is explained by a mismatch between passage-level reranking and document-level qrels evaluation: the LLM reranker scores passages on surface-level query match, while NIST assessors judge document-level relevance.

Hallucination and Failure Cases

Q13 — Corpus coverage failure (all configs). Transmission routes (Q13) produces the lowest NDCG across all configurations (0.316–0.452). Inspection of qrels reveals that Q13's relevant documents include highly specialised epidemiological papers on fomite transmission, aerosol physics, and contact tracing — niche subtopics underrepresented in our random sample of 800 relevant documents. The retrieval system correctly returns what is available; the relevant evidence is simply absent. This is a **corpus coverage failure**, not a retrieval failure. Stratified sampling by query (see Section 4.3) would eliminate this.

Q20 — Medical tone hallucination. The answer to "Are ACE inhibitor patients at increased risk?" states: *"No, patients taking ACE inhibitors are not at increased risk..."* This correctly synthesises the retrieved evidence (multiple studies confirming no elevated risk) but receives LLM score = 1.0 from the judge for directness. However, in a clinical deployment this confident negation without hedging language would be inappropriate. This is a **prompt sensitivity failure**: the grounded-answer prompt encourages confident synthesis, but medical questions require explicit uncertainty quantification. Adding a domain-specific system prompt ("For medical claims, always include appropriate hedging and recommend consulting a physician") would address this.

Q13 transmission routes — Context truncation hallucination. The reranker's generated answer states "airborne transmission is the dominant route for COVID-19." This is a partial truth. The full answer (droplet, contact, fomite, aerosol, and their relative contributions) was present across multiple retrieved chunks, but even with RERANK_TOP_N = 5, the top-ranked passages all focused on airborne evidence, leading to a factually incomplete answer. This is a **ranking homogeneity hallucination**: the reranker over-concentrates on one sub-aspect of the answer, and the generator faithfully but incompletely represents the retrieved evidence.

Query rewriting over-expansion. For Q26 (initial symptoms), the rewriter expanded the query to include "pathophysiology, epidemiology, and diagnostic criteria." This academic broadening

caused retrieval to favour methodology papers over clinical symptom descriptions, hurting MRR (1.0→0.5 in the Hybrid config). Effective biomedical query rewriting must preserve the clinical query type (symptom / treatment / mechanism) rather than adding adjacent academic scope.

Cost Analysis

Configuration	Indexing	Generation	Total	Per Query
Fixed + Small + Vector	\$0.00611	\$0.00515	\$0.01127	\$0.00113
Fixed + Small + Hybrid	\$0.00611	\$0.00504	\$0.01116	\$0.00112
Fixed + Large + Vector	\$0.03974	\$0.00504	\$0.04485	\$0.00449
Sentence + Small + Vector	\$0.00532	\$0.00525	\$0.01058	\$0.00106
Sentence + Small + Hybrid	\$0.00532	\$0.00510	**\$0.01043**	\$0.00104
Sent.+Large+Hybrid+Rerank	\$0.03459	\$0.00304	\$0.03770	\$0.00377

Table 8: Estimated cost per 10-query benchmark run (OpenAI pricing: small embed = \$0.02/1M tokens, large embed = \$0.13/1M tokens, gpt-4o-mini = \$0.15/1M input + \$0.60/1M output tokens). Sentence+Small+Hybrid is the cheapest viable config at \$0.00104/query.

Proposed Improvements Given More Time

1. **Local cross-encoder reranker.** Replace LLM-based reranking with `cross-encoder/ms-marco-MiniLM-L` passage-level relevance scoring at <100ms vs 8–12s per query, zero additional API cost, and better calibrated for document retrieval tasks. Expected NDCG improvement: +5–10 points.
2. **Stratified corpus sampling.** Rather than globally sampling 800 relevant documents, sample proportionally per query, guaranteeing each query has at least 20 relevant documents in the subset. Q13's failure would be eliminated entirely.
3. **Domain-constrained query rewriting.** Add a query-type classifier (symptom / treatment / mechanism / epidemiology) and constrain the rewriter to expand within the same clinical category. This directly addresses the Q26 over-expansion failure.
4. **Parent-document retrieval.** When a chunk matches, return the surrounding paragraph or section. Particularly valuable for biomedical queries where the key claim is often embedded mid-paper rather than in an abstract.
5. **Full BEIR evaluation.** Running across all 18 BEIR datasets (MS-MARCO, HotpotQA, SciFact, NQ, etc.) would validate whether findings generalise beyond the biomedical domain. Our current finding that small embeddings outperform large may be domain-specific.
6. **Structured hallucination detection.** Post-generation, verify each factual claim against retrieved passages using a Natural Language Inference (NLI) model. Flag answers where the claim is not entailed by any retrieved chunk and return "insufficient evidence" rather than a potentially fabricated answer.

Bonus Features Implemented

All bonus features listed in the problem statement were implemented:

Feature	Implementation and Measured Impact	Result
Hybrid Retrieval	RRF fusion of FAISS cosine + BM25 Okapi scores (`hybrid_search()`). Query-dependent: +0.4 NDCG average; improves Q6 (+0.33 P@3), hurts Q26 (MRR 1.0→0.5).	±
LLM Reranking	`gpt-4o-mini` scores each passage 0–10 for relevance. With RERANK_TOP_N=5: NDCG 0.470, P@5 0.820. Identified mismatch: passage-level scoring vs document-level qrels.	∼
Query Rewriting	`gpt-4o-mini` expands queries with synonyms and explicit intent. Domain-specific failure identified: over-expansion hurts clinical precision.	±
Latency analysis	Per-query wall-clock timing for retrieval and generation phases separately.	✓
Cost analysis	Per-run token counting + OpenAI pricing model in `estimate_cost_usd()`. Identifies Sentence+Small+Hybrid as cheapest viable option.	✓
NDCG@10	Graded relevance metric using official NIST scores; not required by PS but standard in TREC evaluations; added for credibility.	✓
Official qrels	NIST human assessments replace self-defined ground truth; metrics directly comparable to published IR literature.	✓

Table 9: Bonus features. ✓= clear positive result, ∼ = marginal / mixed, ± = query-dependent.

Deliverables submitted: (1) Working RAG prototype — `rag_system.py`, `run.py` (2) Dataset description — Section 1; TREC-COVID via `prepare_dataset.py` (3) Evaluation results — Section 3 with tables and per-query analysis (4) Discussion of trade-offs, hallucinations, improvements — Section 4

Models: `text-embedding-3-small`, `text-embedding-3-large`, `gpt-4o-mini` **Libraries:** `faiss-cpu`, `rank-bm25`, `openai`, `numpy` **Dataset:** TREC-COVID (BEIR), 1,300-doc subset, NIST qrels